

Vite React Routing Guide

Vite doesn't have a built-in router because it's a build tool, not a framework. To use routing, you install a library like **React Router**.

Installation

Run this command in your terminal:

```
npm install react-router-dom
```

Recommended File Structure

A standard way to organize your Vite project for routing:

```
my-vite-app/  
├── src/  
│   ├── components/    # Reusable UI (Navbar.jsx)  
│   ├── layouts/        # Page wrappers (RootLayout.jsx)  
│   ├── pages/          # Your views (Home.jsx, About.jsx)  
│   ├── App.jsx         # Component-based routes  
│   └── main.jsx        # Entry point & Data Router setup  
├── index.html          # HTML mount point  
└── package.json
```

Implementation Patterns

1. Modern Standard (Data Router)

Used for better performance by fetching data before the page renders.

main.jsx

```
import { createBrowserRouter, RouterProvider } from "react-router-dom";  
import Home from "../pages/Home";  
  
const router = createBrowserRouter([  
  {  
    path: "/",  
    element: <Home />,  
    loader: async () => fetch("/api/data"), // Pre-fetches data  
  },  
  {
```

```

      path: "/dashboard",
      element: <Dashboard />,
    },
  ]);

ReactDOM.createRoot(document.getElementById("root")).render(
  <RouterProvider router={router} />
);

```

2. Quick Start (Component-Based)

The traditional way to wrap your app and define routes.

main.jsx

```

import { BrowserRouter } from "react-router-dom";

ReactDOM.createRoot(document.getElementById("root")).render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);

```

App.jsx

```

import { Routes, Route, Link } from "react-router-dom";

function App() {
  return (
    <div className="App">
      <nav>
        <Link to="/">Home</Link> | <Link to="/about">About</Link>
      </nav>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
        <Route path="*" element={<h1>404 Not Found</h1>} />
      </Routes>
    </div>
  );
}

```

Why use this with Vite?

- **No full page reloads:** Using <Link> keeps the app fast.

- **HMR:** Vite updates your pages instantly while you code without losing your place in the app.
- **Separation of concerns:** Vite builds the code, React Router handles the URL.